Kaya: one twin per patient

It shows what glucose does to the liver, eyes and kidneys, and lets you test a change before trying it on the real patient.

Open + synthetic data only · Decision support

1 WHAT GOES IN

- Wearable stream
- EHR and labs
- Retina scan
- Patient quick log
- Voice or chat

2 THE KAYA TWIN

Patient state encoder
one fused twin

- **Glucose** — 2-hour forecast and alerts
- **Liver** — fat and fibrosis risk
- **Eye** — retinopathy grade and path
- **Kidney** — eGFR path and risk
- **Heart watch** — cardiovascular event flag, 72 hours ahead

What-if engine
change a meal, walk, sleep or medicine
see the new glucose curve and organ risk

Two streams fused (wearable + record) · explainable · calibrated uncertainty · safety rules in code · one REST API

3 WHO USES IT, AND HOW

PATIENT — on a phone, in 4 languages

1. **Today** — one next action, and why
2. **Quick log** — medicines, walk, mood
3. **Try a change** — see tonight's curve
4. **Ask** — by voice or text

the log becomes the doctor's adherence diary

same twin, same what-if engine

DOCTOR — triage, understand, decide

1. **Triage** — sorted by risk, with reasons
2. **Full record** — glucose, organs, labs
3. **Heart watch** — a week of nights, one risk
4. **What-if studio** — test before the real patient

every action is written to the audit trail

ADMIN — run it safely

1. **Service health** — uptime, requests, latency
2. **Models and data** — what is running, validated how
3. **Audit trail** — who did what, safety stops
4. **Self-test** — checks, hardening list, FHIR API

Kaya is decision support for clinicians, not a diagnostic device. All patients are synthetic. Models, metrics and frameworks are on the next page.

Team Kaya · IIT Hyderabad · Digital Twin Challenge, Phase 1

How Kaya works, at a glance

Learn from open data, build five AI models fusing a wearable stream with the record, make them trustworthy, then put them in people's hands.

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1 LEARNED FROM

- **Glucose recordings** — real people wearing sensors [1]
- **Health survey data** — real labs and liver scans [2]
- **Retina photos** — including Indian patients [3]
- **Synthetic patients** — 2,000 made-up, privacy-safe [4]

2 FIVE AI MODELS

- **Glucose** — predicts the next 2 hours [5]
- **Liver** — fat and scarring risk [6]
- **Eye** — grades diabetic eye damage [7]
- **Kidney** — risk and 5-year decline [6]
- **Heart watch** — week of nights + record [9]

3 MADE TRUSTWORTHY

- **Says how sure it is** — honest uncertainty bands
- **Shows why** — reasons and heat-maps
- **Safety rules** — emergency and dose checks [8]
- **Private by design** — no real patient data

4 WHAT PEOPLE GET

- **Doctor dashboard** — who needs help, and why
- **Patient app** — plain words, 4 languages, voice [8]
- **Admin page** — health, models, audit trail
- **What-if testing** — try a change on the twin first

BUILT WITH

PyTorch · LightGBM · EfficientNet · SHAP, Grad-CAM · Sarvam AI · FastAPI · React · Docker

FOOTNOTES

the detail behind each icon above

1. **CGMacros (PhysioNet)**
   Real glucose, meals and Fitbit data from 34 people.
2. **NHANES 2017-18**
   4,745 adults with FibroScan liver labels; 4,827 for kidney.
3. **APTOS 2019 + IDRiD**
   Retina photographs: APTOS for training, Indian IDRiD for testing.
4. **Synthetic cohort**
   Our generators + simglucose: 2,000 patients; a coupled 28-day cohort links sleep, HRV and glucose.
5. **Glucose model**
   GRU (PyTorch) + LightGBM ensemble, conformal 80% bands, isotonic-calibrated alerts.
   RMSE 12.6 / 21.1 / 28.2 mg/dL at 30 / 60 / 120 min; catches 72% of lows, 45 min ahead.
6. **Liver and kidney models**
   LightGBM + logistic regression, explained with SHAP.
   AUROC: liver 0.83 (standard FIB-4: 0.63), kidney 0.85. 1.0 is perfect, 0.5 a coin flip.
7. **Eye model**
   EfficientNet-B2 (timm), grades 0-4, Grad-CAM heat-maps. Kappa 0.89 on APTOS, 0.71 on IDRiD.
   Eye progression and the 5-year kidney path are simulations.
8. **Voice and documents (Sarvam AI, server-side only)**
   saaras:v4 speech-to-text, sarvam-105b-conversations chat, bulbul:v4-flash text-to-speech.
   Sarvam Document AI reads uploaded reports. Emergency and dose rules run in code, before the model.
9. **Heart watch (two streams, fused)**
   Wearable week (LightGBM) + record (logistic), combined by a stacker. AUROC 0.83; catches 76% of cardiovascular events a median 23 h ahead at about 1 false alert per patient-month. Synthetic data.

Good to know
Live mode runs the GRU in pure NumPy: about 1 ms per reading, no GPU.
Not built yet: consent records, a persistent audit log, ABDM. Not clinically validated.